

Unit 1 Extended Practice Packet

Foundations of Algebra

Slope • Forms & Intercepts • Writing Equations • Graphing • Applications • Arithmetic Sequences • Functions • Key Features

Key Ideas

Slope: $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{rise}}{\text{run}}$

Horizontal line: $m = 0$; vertical: undefined

Slope-intercept: $y = mx + b$

Point-slope: $y - y_1 = m(x - x_1)$

Standard form: $Ax + By = C$

x-intercept: let $y = 0$, solve for x

y-intercept: let $x = 0$, solve for y

Explicit rule: $a_n = a_1 + d(n - 1)$

Recursive rule: $a_1 = \text{first term}$, $a_n = a_{n-1} + d$

Sequences are *discrete*: plot dots, don't connect.

Domain = inputs (x -values)

Range = outputs (y -values)

$f(3)$ means "substitute $x = 3$ "

$[]$ includes ($\leq, \geq$); $()$ excludes ($<, >$)

∞ and $-\infty$ always get $()$

Worked Example

Graph the line $2x + 3y = 12$. Find both intercepts, rewrite the equation in slope-intercept form, and check with a table.

Step 1 – x-intercept (let $y = 0$):

$$2x + 3(0) = 12 \Rightarrow 2x = 12 \Rightarrow x = 6 \quad (6, 0)$$

Step 2 – y-intercept (let $x = 0$):

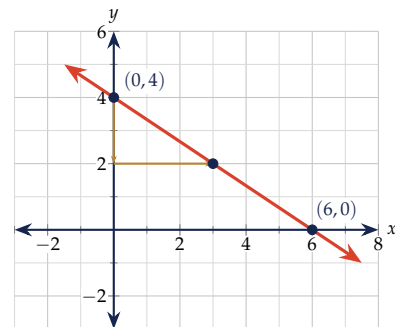
$$2(0) + 3y = 12 \Rightarrow 3y = 12 \Rightarrow y = 4 \quad (0, 4)$$

Step 3 – Slope-intercept form. Subtract $2x$, then divide by 3:

$$3y = -2x + 12 \Rightarrow y = -\frac{2}{3}x + 4$$

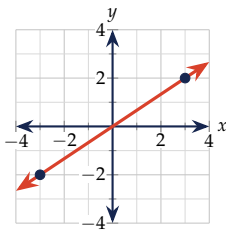
Step 4 – Graph. Plot both intercepts and draw the line. Check with the slope $-\frac{2}{3}$: from $(0, 4)$ go *down 2, right 3* to $(3, 2)$ — it lands on the line.

x	0	3	6
y	4	2	0



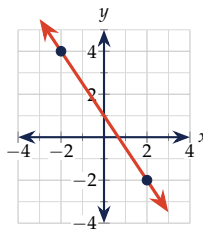
Part A – Slope from Graphs, Points & Tables

1. Find the slope of the line.



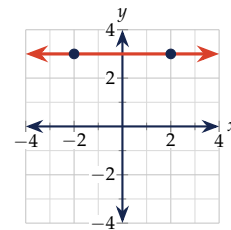
$m =$ _____

2. Find the slope of the line.



$m =$ _____

3. Find the slope of the line.



$m =$ _____

4. Slope through
(1, -3) and (4, 9).

$m =$ _____

5. Slope through
(-6, 2) and (3, -4).

$m =$ _____

6. Slope through
(-2, 7) and (5, 7).

$m =$ _____

7. Slope through
(3, 1) and (3, -6).

$m =$ _____

8. Find the slope from the table.

x	0	3	6	9
y	-4	-2	0	2

$m =$ _____

9. Find the slope from the table.

x	-2	-1	0	1
y	10	7	4	1

$m =$ _____

10. Find the slope from the table.

x	5	10	15	20
y	12	22	32	42

$m =$ _____

Part B – Slope-Intercept Form, Intercepts & Converting Forms

11. Slope and y -intercept of

$$y = 7x - 3$$

$m =$ _____ $b =$ _____

12. Slope and y -intercept of

$$y = -x + 8$$

$m =$ _____ $b =$ _____

13. Slope and y -intercept of

$$y = \frac{2}{5}x$$

$m =$ _____ $b =$ _____

14. Find the x -intercept of $4x + 3y = 24$.

Answer: _____

15. Find the y -intercept of $2x - 7y = 14$.

Answer: _____

16. Find both intercepts of $-5x + 2y = 10$.

x -int: _____ y -int: _____

17. Find both intercepts of $6x - 9y = 36$.

x -int: _____ y -int: _____

Rewrite each equation in slope-intercept form $y = mx + b$.

18.

$$3x + y = 7$$

Answer: _____

19.

$$8x + 4y = 20$$

Answer: _____

20.

$$-10x - 5y = 15$$

Answer: _____

21.

$$2x - 3y = 9$$

Answer: _____

22.

$$x + 4y = -8$$

Answer: _____

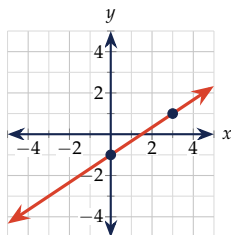
23.

$$-4x + 6y = 18$$

Answer: _____

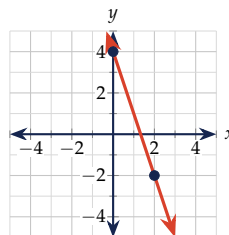
Part C – Writing Equations of Lines

24. Write the equation of the line shown.



Answer: _____

25. Write the equation of the line shown.



Answer: _____

26. Slope 4, through $(1, -2)$.

Answer: _____

27. Slope -2 , through $(-3, 10)$.

Answer: _____

28. Slope $\frac{1}{3}$, through $(6, 5)$.

Answer: _____

29. Through $(1, 3)$ and $(4, 12)$.

Answer: _____

30. Through $(-2, 9)$ and $(3, -1)$.

Answer: _____

31. Through $(5, -2)$ and $(-5, 4)$.

Answer: _____

32. A line has x -intercept $(3, 0)$ and y -intercept $(0, -6)$. Write its equation in slope-intercept form.

Answer: _____

33. Write the equation of the line represented by the table.

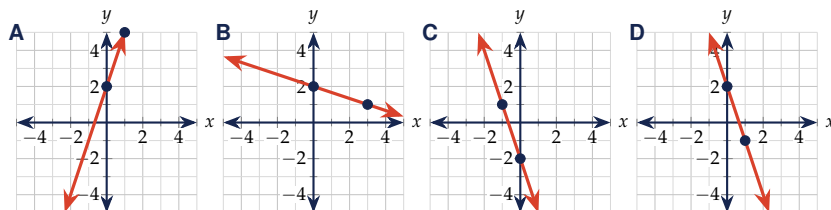
x	0	1	2	3
y	-5	-1	3	7

Answer: _____

Part D – Graphing Linear Equations

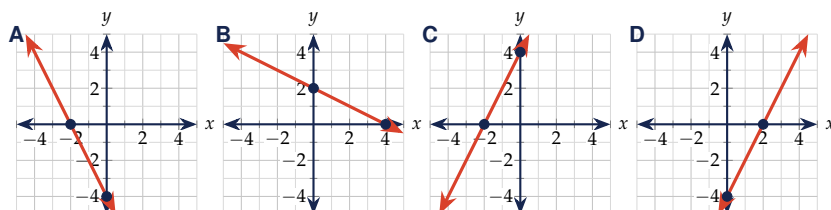
34. Which graph represents $y = -3x + 2$?

Answer: _____



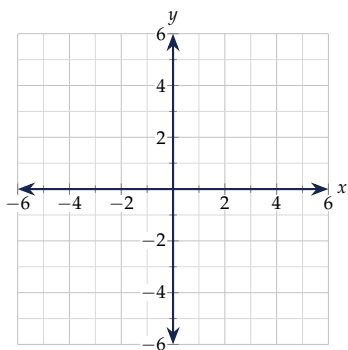
35. A line has x -intercept $(-2, 0)$ and y -intercept $(0, 4)$. Which graph represents the line?

Answer: _____

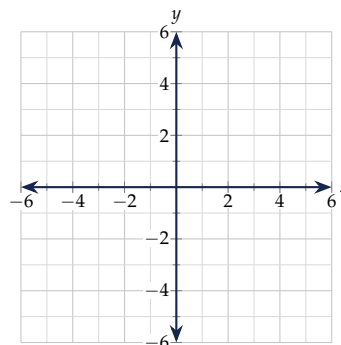


36. Complete the table for $y = -2x + 5$, then graph the line.

x	0	1	2	3
y				



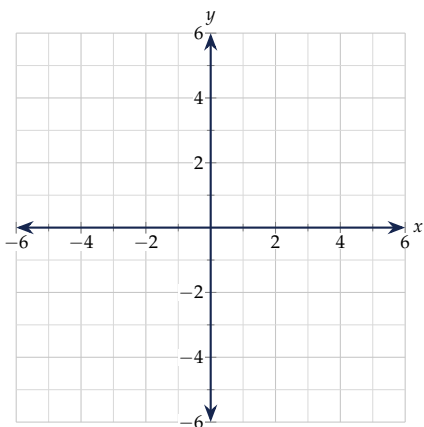
37. Graph the horizontal line $y = -2$ and the vertical line $x = 3$ on the same grid. Where do they intersect?



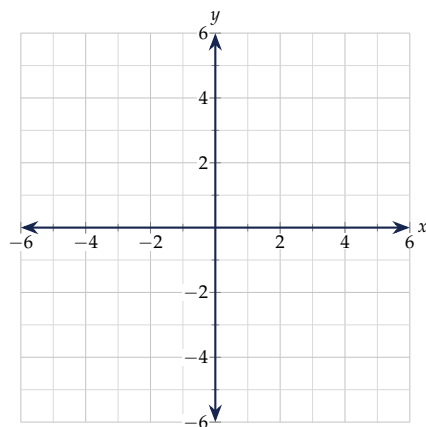
Answer: _____

Graph each line on the grid. Show your method (slope-intercept or intercepts).

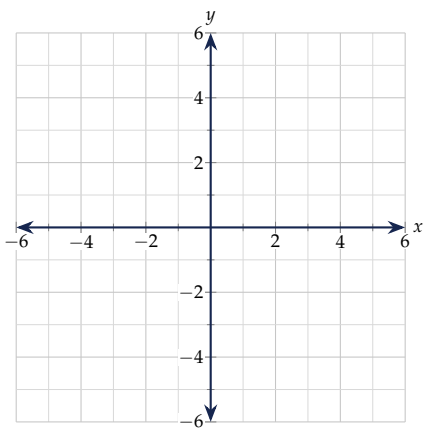
38. $y = \frac{1}{2}x - 3$



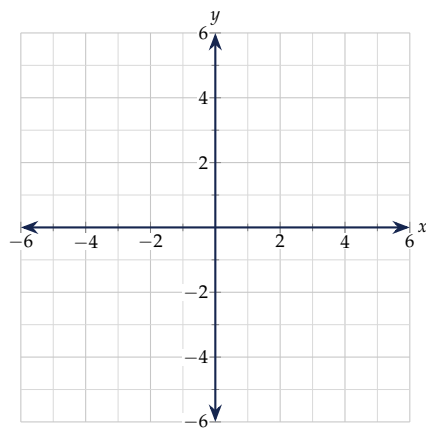
39. $y = -\frac{4}{3}x + 2$



40. $4x + 2y = 8$



41. $3x - 5y = 15$



42. A streaming service charges a \$15 setup fee plus \$12 per month, modeled by $y = 12x + 15$. After how many months will the total cost reach \$123?

Answer: _____

43. A taxi charges \$4 to start the ride plus \$2.50 per mile. How many miles can you ride for \$29?

Answer: _____

44. An electrician charges \$65 per hour plus a flat service fee. A 3-hour job costs \$245.

(a) Write an equation for the total cost y of an x -hour job.

Answer: _____

(b) Find the cost of a 5-hour job.

Answer: _____

45. A moving company charges \$30 per hour plus a flat fee. A 6-hour move costs \$210.

(a) Write an equation for the total cost y of an x -hour move.

Answer: _____

(b) Find the cost of a 9-hour move.

Answer: _____

46. A caterer charges \$170 for 10 guests and \$290 for 25 guests. The cost is linear.

(a) Find the cost per guest (the slope).

Answer: _____

(b) Write an equation for the cost y of x guests.

Answer: _____

(c) How much would 40 guests cost?

Answer: _____

47. A candle is 30 cm tall and burns down 2.5 cm every hour.

(a) Write an equation for the height y after x hours.

Answer: _____

(b) What does the y -intercept mean in this situation?

(c) After how many hours will the candle burn out?

Answer: _____

48. At the bowling alley, Andre pays \$4 to rent shoes and \$5 per game. How many games can he bowl with \$34?

Answer: _____

49. The table shows the cost of renting a kayak. Write an equation for the cost y of x hours.

Hours x	1	2	3	4
Cost y	\$22	\$34	\$46	\$58

Answer: _____

Part F – Arithmetic Sequences

50. Is each sequence arithmetic? If yes, give the common difference d .

(a) 4, 9, 14, 19, ...

(b) 2, 4, 8, 16, ...

(c) 20, 17, 14, 11, ...

(d) 1, 4, 9, 16, ...

51. The 3rd term of an arithmetic sequence is 10 and $d = 6$. Find the 15th term.

Answer: _____

52. The 8th term of an arithmetic sequence is 40 and $d = -5$. Find the 20th term.

Answer: _____

53. Find the 60th term of 7, 11, 15, 19, ...

Answer: _____

54. Find the 30th term of 25, 22, 19, 16, ...

Answer: _____

55. Write an *explicit* rule for -6, -1, 4, 9, ...

$a_n =$ _____

56. Write a *recursive* rule for 40, 33, 26, 19, ...

$a_1 =$ _____, $a_n =$ _____

57. Find the first 5 terms:

$$a_1 = 2, \quad a_n = a_{n-1} + 6$$

Answer: _____

58. Find the first 5 terms:

$$a_1 = 50, \quad a_n = a_{n-1} - 12$$

Answer: _____

59. Which is the recursive rule for 6, 15, 24, 33, ...?

A) $a_1 = 6, \quad a_n = a_{n-1} + 9$

B) $a_1 = 6, \quad a_n = 9a_{n-1}$

C) $a_n = 9 + 6(n - 1)$

D) $a_1 = 9, \quad a_n = a_{n-1} + 6$

Answer: _____

60. In the sequence 5, 12, 19, 26, ... which term number is 124?

Hint: set $a_n = 124$ and solve for n .

$n =$ _____

61. Given the sequence 3, 7, 11, 15, 19, ...

(a) Write an explicit rule for the sequence.

Answer: _____

(b) Write a recursive rule for the sequence.

Answer: _____

(c) Find the 75th term.

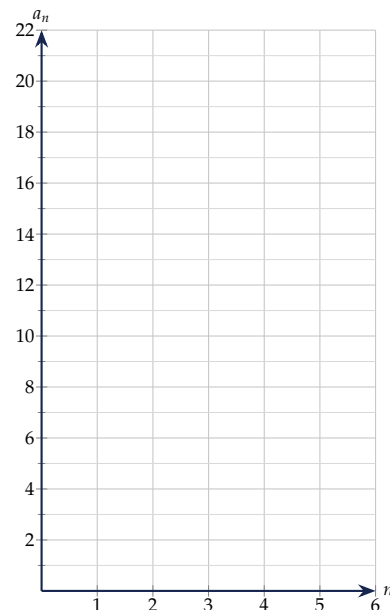
Answer: _____

(d) Complete the table and graph the first 5 terms. Continuous or discrete?

n	1	2	3	4	5
a_n					

(e) Domain of the first 5 terms: _____

Range of the first 5 terms: _____



62. A theater's first row has 18 seats. Each row after that has 3 more seats than the row in front of it.

(a) Write an explicit rule for the number of seats in row n .

Answer: _____

(b) How many seats are in row 20?

Answer: _____

63. Jada saves money each week. She starts with \$35 and adds \$15 every week.

(a) Write an explicit rule for her savings in week n (week 1 = \$35).

Answer: _____

(b) In which week will she have \$230?

Answer: _____

Part G – Functions & Function Notation

64. $f(x) = 5x - 2$, domain $\{-1, 0, 3\}$. Find the range.

Answer: _____

65. $f(x) = -2x + 9$, domain $\{-4, 2, 6\}$. Find the range.

Answer: _____

66. Given $g(x) = 2x^2 - 3$, find $g(4)$.

Answer: _____

67. Given $f(x) = x^2 + 2x - 5$, find $f(3)$.

Answer: _____

68. Given $f(x) = x^2 - 6x + 1$, find $f(-1)$.

Answer: _____

69. Given $f(x) = 3x + 11$, find x when $f(x) = -4$.

Answer: _____

70. Given $f(x) = -4x + 7$, find x when $f(x) = -5$.

Answer: _____

71. Fill in using the word bank.

Word bank: *input, output, x-values, y-values, independent, dependent*

Domain: _____

Range: _____

72. Is each relation a function? Write **Yes** or **No** and explain.

(a) $\{(1, 2), (2, 4), (3, 6)\}$

(b) $\{(2, 5), (2, 7), (4, 1)\}$

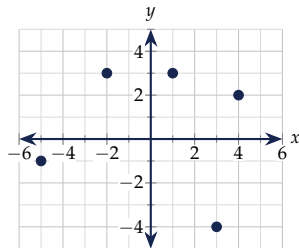
(c) $\{(-1, 3), (0, 3), (1, 3)\}$

Part H – Domain, Range, Interval Notation & Key Features

73. Find the domain and range of the discrete points.

Domain: _____

Range: _____



74. Find the domain and range of the relation.

x	-3	-1	2	5
y	4	0	4	-6

Domain: _____

Range: _____

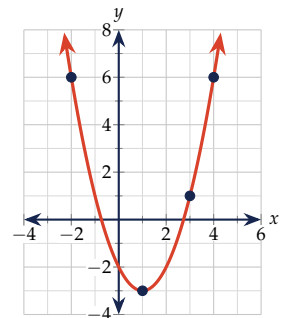
75. Use the graph of f to evaluate.

(a) $f(1) =$ _____

(b) $f(3) =$ _____

(c) $f(0) =$ _____

(d) If $f(x) = 6$, then $x =$ _____ or $x =$ _____



76. Convert to interval notation.

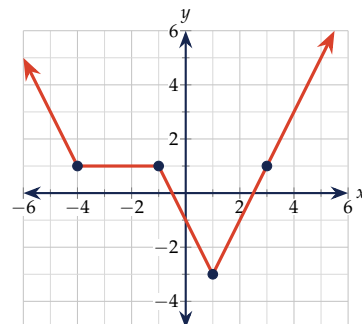
- (a) $x \geq 4$ _____ (b) $x < -1$ _____
 (c) $-3 < x \leq 5$ _____ (d) $0 \leq x \leq 8$ _____

77. Convert to inequality notation.

- (a) $(2, \infty)$ _____ (b) $(-\infty, 6]$ _____
 (c) $[-7, -1)$ _____ (d) $(-2, 4)$ _____

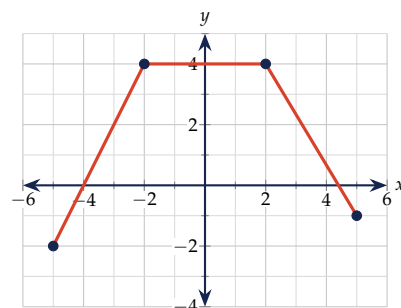
78. Use the graph of f to answer each question. (Arrows mean the graph keeps going.)

- (a) $f(-4) =$ _____ (b) $f(3) =$ _____
 (c) Domain: _____ (d) Range: _____
 (e) Interval(s) of increase: _____
 (f) Interval(s) of decrease: _____
 (g) Constant interval: _____
 (h) As $x \rightarrow \infty$, $y \rightarrow$ _____ As $x \rightarrow -\infty$, $y \rightarrow$ _____



79. Use the graph of f . (Closed dots are endpoints.)

- (a) $f(0) =$ _____ (b) $f(-5) =$ _____
 (c) Domain (inequality and interval): _____
 (d) Range (inequality and interval): _____
 (e) Increasing: _____ Decreasing: _____
 (f) Constant: _____ x -intercept on the left: _____



80. Find the key features of the graph. (Closed dot = endpoint; arrow = keeps going.)

Domain:	Range:
x -intercept:	y -intercept:
Interval of increase:	Interval of decrease:
End behavior: as $x \rightarrow \infty$, $y \rightarrow$ _____	

